

Problem Set 8
Due: Apr. 26, 2021

1. Section 4.1 Problem 4.

The metro train arrives at the station near your home every quarter hour starting at 6:00 a.m. You walk into the station every morning between 7:10 and 7:30 a.m., with the time in this interval being a random variable with given PDF (cf. Example 3.14, in Chapter 3). Let X be the elapsed time, in minutes, between 7:10 and the time of your arrival. Let Y be the time that you have to wait until you board a train. Calculate the CDF of Y in terms of the CDF of X and differentiate to obtain a formula for the PDF of Y .

Solution: If you arrive before 7:15 a.m., you will board the 7:15 train, or you will board the 7:30 train. Therefore, we have

$$F_Y(y) = \begin{cases} 0 & \text{if } y \leq 0 \\ \mathbf{P}(5 - y \leq X \leq 5) + \mathbf{P}(20 - y \leq X \leq 20) & \text{if } 0 < y \leq 5 \\ \mathbf{P}(20 - y \leq X \leq 20) & \text{if } 5 < y \leq 15 \\ 1 & \text{if } y > 15 \end{cases}$$

Using the CDF of X , we have

$$\mathbf{P}(5 - y \leq X \leq 5) = F_X(5) - F_X(5 - y)$$

$$\mathbf{P}(20 - y \leq X \leq 20) = F_X(20) - F_X(20 - y)$$

Thus,

$$F_Y(y) = \begin{cases} 0 & \text{if } y \leq 0 \\ F_X(5) - F_X(5 - y) + F_X(20) - F_X(20 - y) & \text{if } 0 < y \leq 5 \\ F_X(20) - F_X(20 - y) & \text{if } 5 < y \leq 15 \\ 1 & \text{if } y > 15 \end{cases}$$

Differentiating, we obtain

$$f_Y(y) = \begin{cases} f_X(5 - y) + f_X(20 - y) & \text{if } 0 < y \leq 5 \\ f_X(20 - y) & \text{if } 5 < y \leq 15 \\ 0 & \text{otherwise} \end{cases}$$

2. Section 4.1 Problem 9.

Consider the same problem as in Example 4.9, but assume that the random variables X and Y are independent and exponentially distributed with different parameters λ and μ , respectively. Find the PDF of $X - Y$.

Solution: Let $Z = X - Y$. We will first calculate the CDF $F_Z(z)$ by considering separately the cases $z \geq 0$ and $z < 0$. For $z \geq 0$, we have (see the left side of Fig. 4.6)

$$\begin{aligned}
 F_Z(z) &= \mathbf{P}(X - Y \leq z) \\
 &= 1 - \mathbf{P}(X - Y > z) \\
 &= 1 - \int_0^\infty \left(\int_{z+y}^\infty f_{X,Y}(x, y) dx \right) dy \\
 &= 1 - \int_0^\infty \mu e^{-\mu y} \left(\int_{z+y}^\infty \lambda e^{-\lambda x} dx \right) dy \\
 &= 1 - \int_0^\infty \mu e^{-\mu y} e^{-\lambda(z+y)} dy \\
 &= 1 - e^{-\lambda z} \int_0^\infty \mu e^{-(\lambda+\mu)y} dy \\
 &= 1 - \frac{\mu}{\lambda + \mu} e^{-\lambda z}
 \end{aligned}$$

For the case $z < 0$,

$$\begin{aligned}
 F_Z(z) &= \mathbf{P}(X - Y \leq z) \\
 &= \int_0^\infty \left(\int_{x-z}^\infty f_{X,Y}(x, y) dy \right) dx \\
 &= \int_0^\infty \lambda e^{-\lambda x} \left(\int_{x-z}^\infty \mu e^{-\mu y} dy \right) dx \\
 &= \int_0^\infty \lambda e^{-\lambda x} e^{-\mu(x-z)} dx \\
 &= \frac{\lambda}{\lambda + \mu} e^{\mu z}
 \end{aligned}$$

Combining the two cases, we obtain

$$F_Z(z) = \begin{cases} 1 - \frac{\mu}{\lambda + \mu} e^{-\lambda z} & \text{if } z \geq 0 \\ \frac{\lambda}{\lambda + \mu} e^{\mu z} & \text{if } z < 0 \end{cases}$$

The PDF of Z is obtained by differentiating its CDF. We have

$$f_Z(z) = \begin{cases} \frac{\lambda\mu}{\lambda + \mu} e^{-\lambda z} & \text{if } z \geq 0 \\ \frac{\lambda\mu}{\lambda + \mu} e^{\mu z} & \text{if } z < 0 \end{cases}$$

3. Find the probability density function of $Y = e^X$ when X is normally distributed with parameters μ and σ^2 . The random variable Y is said to have a *lognormal distribution* (since $\log Y$ has a normal distribution) with parameters μ and σ^2 .

Solution:

$$F_Y(y) = \mathbf{P}(e^X \leq y) = \mathbf{P}(X \leq \log y) = F_X(\log y)$$

$$f_Y(y) = \frac{d}{dy} F_X(\log y) = \frac{f_X(\log y)}{y} = \frac{1}{y\sqrt{2\pi}\sigma} e^{-(\log y - \mu)^2 / (2\sigma^2)}$$

4. A point (X, Y) is picked at random uniformly in the unit circle. Find the joint PDF of R and X , where $R^2 = X^2 + Y^2$.

Solution:

$$\begin{aligned} F_{R,X}(r, x) &= \mathbf{P}(R \leq r, X \leq x) \\ &= \int_{-r}^x \left(\int_{-\sqrt{r^2-u^2}}^{\sqrt{r^2-u^2}} f_{X,Y}(u, v) dv \right) du \\ &= \int_{-r}^x \frac{1}{\pi} \cdot 2\sqrt{r^2-u^2} du \end{aligned}$$

The integral region is the left part of the unit circle separated by a vertical line $u = x$.

$$f_{R,X}(r, x) = \frac{\partial^2}{\partial r \partial x} F_{R,X}(r, x) = \frac{2r}{\pi\sqrt{r^2-x^2}}, \quad |x| < r < 1$$